

Statistical and Dimensionality-Reduction Analysis of the Communities and Crime Dataset

Serkan Eren¹

¹Artificial Intelligence Graduate Program, Department of Computer Engineering, Faculty of Engineering, Gebze Technical University, Kocaeli, Türkiye
s.eren2018@gtu.edu.tr

This work was conducted as the final project of the CSE 555: Statistical Pattern Recognition course in the Master's Degree Program at Gebze Technical University.

ABSTRACT This work investigates the structure of the UCI *Communities and Crime* dataset through a sequence of exploratory, statistical, and dimensionality-reduction analyses. A curated subset of eighteen normalized sociodemographic variables is studied together with a three-level discretization of the continuous target *ViolentCrimesPerPop* into Low, Medium, and High crime classes by tertiles. The pipeline includes per-feature box-plot inspection and IQR-based outlier cleaning, Pearson correlation analysis, z-score normalization with per-feature Fisher Distance, principal component analysis (PCA) with Fisher Distance per principal component, linear discriminant analysis (LDA), and a comparison of K-Means, DBSCAN, t-SNE, and Self-Organizing Map projections. Quantitative separability is reported through silhouette and 5-NN cross-validation accuracy on the 2-D embeddings. The analysis confirms that family-structure and socioeconomic variables (e.g., *PctKids2Par*, *PctIlleg*, *PctPopUnderPov*) carry the strongest individual class signal, that the leading PCA components inherit most of this discriminative content, and that LDA produces a tighter class-separable 2-D embedding than PCA. A Welch's t-test on *PctKids2Par* between Low and High classes confirms a highly significant mean difference at both $n = 36$ and $n = 64$.

INDEX TERMS Communities and Crime, exploratory data analysis, Fisher Distance, PCA, LDA, K-Means, DBSCAN, t-SNE, Self-Organizing Map, hypothesis testing.

I. INTRODUCTION

THE *Communities and Crime* dataset from the UCI Machine Learning Repository [1] is a widely used benchmark for sociodemographic modelling of community-level violent crime in the United States. It aggregates information from the 1990 U.S. Census, the 1990 U.S. Law Enforcement Management and Administrative Statistics (LEMAS) survey, and the 1995 FBI Uniform Crime Reports (UCR), and contains 1 994 communities described by 128 attributes. Each predictive attribute is already normalized to the $[0, 1]$ interval by the dataset authors so that comparisons across communities of different population are meaningful.

The aim of this study is not to build a competitive predictive model, but to apply the statistical and pattern-recognition tools taught in CSE 555 to a real, mixed-quality dataset, and to interpret what each tool exposes about the data. To reduce dimensionality to a tractable yet representative working set, eighteen sociodemographic variables covering age structure, ethnic composition, education, employment, family stability, housing, and immigration are selected (Table 1). To obtain a

multi-class problem as required by the assignment, the continuous response *ViolentCrimesPerPop* is discretized into three equal-frequency classes *Low*, *Medium*, and *High* by its tertiles.

The remainder of the paper is organized as follows. Section II describes the dataset and the preprocessing pipeline. Sections III–X address the eight analytical tasks of the project in order: distributions and outliers (Section III), correlation analysis (Section IV), per-feature Fisher Distance after z-scoring (Section V), PCA-based Fisher analysis (Section VI), top/bottom PC scatter plots (Section VII), LDA (Section VIII), clustering with K-Means, DBSCAN, t-SNE, and SOM (Section IX), and a Welch t-test (Section X). Section XI summarizes the main findings.

II. DATASET AND PREPROCESSING

The Communities and Crime dataset [1] contains 1 994 rows and 128 columns, of which five are non-predictive identifiers (*state*, *county*, *community*, *communityname*, *fold*). A non-trivial fraction of the LEMAS-derived

TABLE 1. Selected features and their interpretation. All values are normalized to [0, 1] by the dataset authors.

Feature	Description
population	Community population
householdsize	Mean household size
racepctblack	Percent of Black population
racePctWhite	Percent of White population
agePct12t29	Percent of population aged 12–29
agePct65up	Percent of population aged ≥ 65
pctUrban	Percent of population living in urban areas
medIncome	Median household income
PctPopUnderPov	Percent of population under poverty line
PctNotHSGrad	Percent without a high-school diploma
PctBSorMore	Percent with a Bachelor's degree or higher
PctUnemployed	Percent unemployed
MalePctDivorce	Percent of divorced males
PctKids2Par	Percent of kids in two-parent households
PctIlleg	Percent of children born to never-married parents
NumImmig	Number of foreign-born residents
PctHousOwnOcc	Percent of owner-occupied housing units
PctVacantBoarded	Percent of boarded-up vacant housing

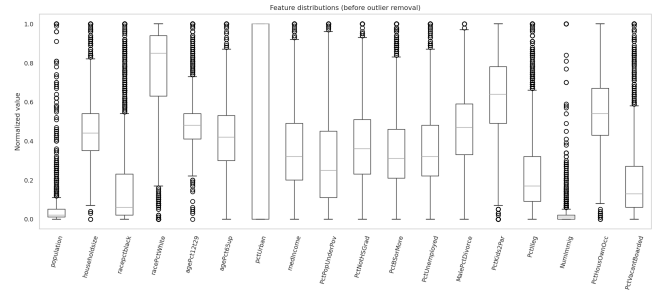
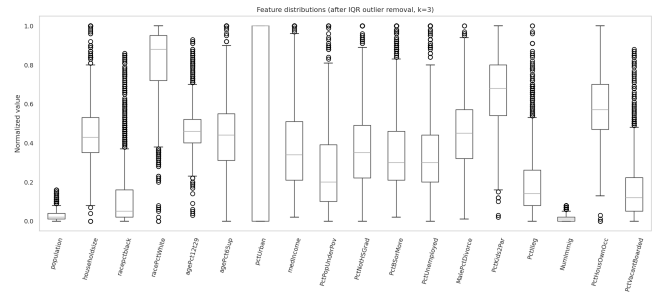
columns is missing for most communities; those columns were therefore excluded from the working subset. After identifier removal, a curated set of 18 sociodemographic features (Table 1) was selected based on (i) low missingness, (ii) interpretability for crime-related analysis, and (iii) coverage of distinct conceptual domains (demographics, economy, education, family, housing, immigration). The continuous response *ViolentCrimesPerPop* was discretized by tertiles into three classes *Low*, *Medium*, and *High* (679/662/653 samples before cleaning).

Any residual missing values in the working subset were imputed by the per-column median. All subsequent analyses were carried out in Python 3.10 using NumPy, pandas, scikit-learn, scipy, seaborn, and MiniSom. The complete Jupyter notebook accompanies this report.

III. DISTRIBUTIONS AND OUTLIER DETECTION (TASK 1)

Figure 1 presents per-feature box plots of the eighteen selected variables before outlier removal. Several features are visibly right-skewed and exhibit a long upper tail of extreme values — most notably *population*, *NumImmig*, *PctIlleg*, *PctVacantBoarded*, and *PctPopUnderPov*. These tails reflect a small set of communities that are demographically or socioeconomically extreme (very large cities, communities with very high immigration counts, or extreme deprivation). In contrast, variables such as *agePct12t29*, *agePct65up*, and *householdsize* are tightly concentrated around the population median with comparatively few outliers.

A multivariate IQR-based filter was applied as the outlier detection method. For each feature the inter-quartile range $IQR = Q_3 - Q_1$ was computed, and a sample was retained only if all of its feature values lay within $[Q_1 - k \cdot IQR, Q_3 + k \cdot IQR]$. A conservative multiplier $k = 3$ (rather than

**FIGURE 1.** Box plots of the eighteen selected features prior to outlier removal. Several socioeconomic variables show clear right-skew with long upper tails.**FIGURE 2.** Box plots after IQR-based cleaning ($k = 3$). The most extreme tails are removed while the bulk of each distribution is preserved.

the textbook 1.5) was chosen so that genuinely extreme but informative observations are kept and only the most severe outliers are removed. With $k = 3$ the filter discards 358 rows (18.0%), leaving 1 636 communities with class distribution *Low*: 647, *Medium*: 571, *High*: 418.

An important interpretive observation is that the *High*-crime class shrinks the most (from 653 to 418), confirming that high-crime communities tend to occupy the extreme regions of multiple feature distributions simultaneously (high *PctPopUnderPov*, high *PctIlleg*, high *PctVacantBoarded*, low *PctKids2Par*). Outlier removal therefore systematically reduces class imbalance toward the centre of feature space; this is taken into account when interpreting the separability of the cleaned data in later sections. All subsequent analyses use the cleaned dataset.

IV. CORRELATION ANALYSIS (TASK 2)

The full Pearson correlation matrix among the eighteen features and the ordinally encoded class label is shown in Figure 3. Several intuitive sociodemographic block structures emerge. Income, education, family stability, and home ownership form a positively correlated block (*medIncome*, *PctBSorMore*, *PctKids2Par*, *PctHousOwnOcc*), while indicators of deprivation form a separate positively correlated block (*PctPopUnderPov*, *PctNotHSGrad*, *PctUnemployed*, *PctVacantBoarded*, *PctIlleg*, *MalePctDivorce*). The two blocks are strongly anti-correlated with each other, reflecting the well-known socioeconomic gradient.

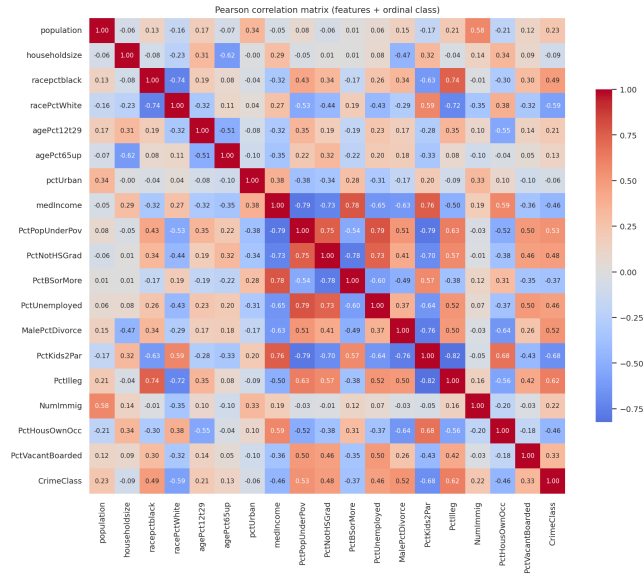


FIGURE 3. Pearson correlation matrix of the eighteen features and the ordinal crime class.



FIGURE 4. Per-feature Pearson correlation with the (ordinal) crime class. Red bars indicate positive correlation with higher crime levels.

Figure 4 ranks features by absolute correlation with the (ordinal) crime class. The strongest individual correlates are PctKids2Par (-0.68), PctIlleg ($+0.62$), racePctWhite (-0.59), PctPopUnderPov ($+0.54$), MalePctDivorce ($+0.52$), and medIncome (-0.46). The signs are consistent with the established sociological literature on community-level crime: family stability, education, income, and home ownership are protective, while poverty, family disruption, unemployment, and housing decay co-occur with higher violent crime. Demographic-only variables such as pctUrban, householdsize, agePct12t29, and agePct65up carry very little individual class signal.

V. Z-SCORE NORMALIZATION AND PER-FEATURE FISHER DISTANCE (TASK 3)

After z-scoring, the per-feature Fisher Distance was computed as a multi-class generalisation of the classical two-class

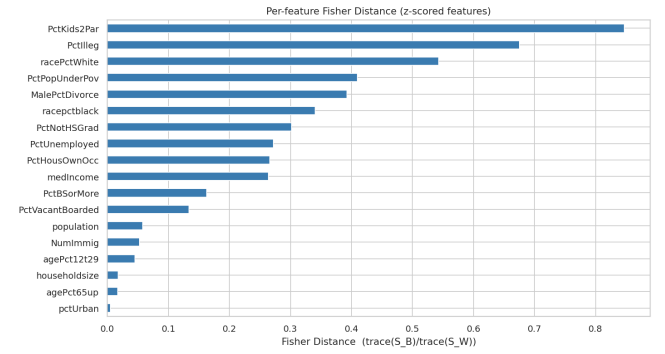


FIGURE 5. Per-feature Fisher Distance after z-score normalization. PctKids2Par and PctIlleg dominate.

Fisher score:

$$J(f) = \frac{\sum_{c=1}^C n_c (\mu_c^{(f)} - \mu^{(f)})^2}{\sum_{c=1}^C \sum_{i \in c} (x_i^{(f)} - \mu_c^{(f)})^2}, \quad (1)$$

where $\mu_c^{(f)}$ is the class- c mean of feature f and $\mu^{(f)}$ is its global mean. Larger $J(f)$ indicates that feature f separates the class means well relative to within-class dispersion.

The resulting ranking (Figure 5) is largely consistent with the correlation ranking of Section IV. The two top-ranked features are PctKids2Par ($J=0.85$) and PctIlleg ($J=0.68$), followed by racePctWhite, PctPopUnderPov, and MalePctDivorce. Features such as pctUrban, householdsize, and agePct65up have Fisher Distance values close to zero, indicating that their class-conditional distributions are nearly indistinguishable. The Fisher Distance and absolute class correlation rankings agree well, since for univariate normalized features both quantities effectively measure the contrast between class means relative to within-class scatter.

VI. PCA AND FISHER DISTANCE PER PRINCIPAL COMPONENT (TASK 4)

PCA was applied to the z-scored data, producing 18 principal components. The leading component PC1 alone explains 39.6% of the variance, and the first five components together account for 79.4% of the total variance (Figure 6). This is consistent with the strong correlation block-structure observed in Section IV: a single socioeconomic “advantage–disadvantage” axis dominates the feature space.

Treating each PC projection as a feature, the Fisher Distance from Equation (1) was recomputed (Figure 7). The Fisher Distance of PC1 is 0.78, which is the second-highest score in the entire study (only the original PctKids2Par feature scores higher). All other principal components have Fisher Distance below 0.05.

There is a strong positive relationship between the eigenvalue of a principal component (its variance) and its Fisher Distance: the Pearson correlation between the two is $r = 0.93$ and the Spearman rank correlation is $\rho = 0.66$ (Figure 8). The result, however, must be interpreted with care. The high

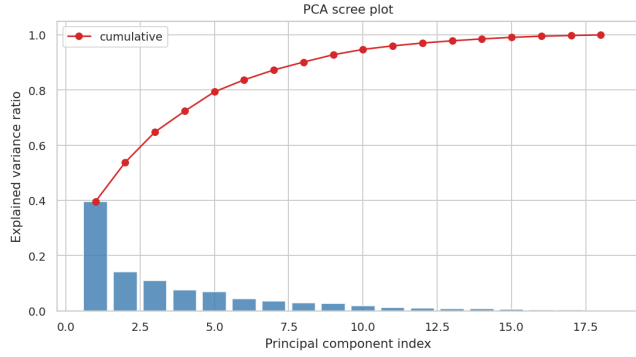


FIGURE 6. PCA scree plot. PC1 captures 39.6% of the variance; the top five components capture 79.4%.

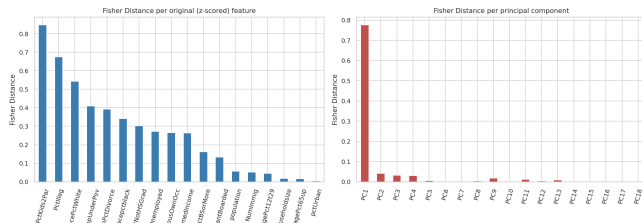


FIGURE 7. Per-feature Fisher Distance for the z-scored original features (left) and for the 18 principal components (right). PC1 inherits most of the discriminative content of the leading correlated block of features.

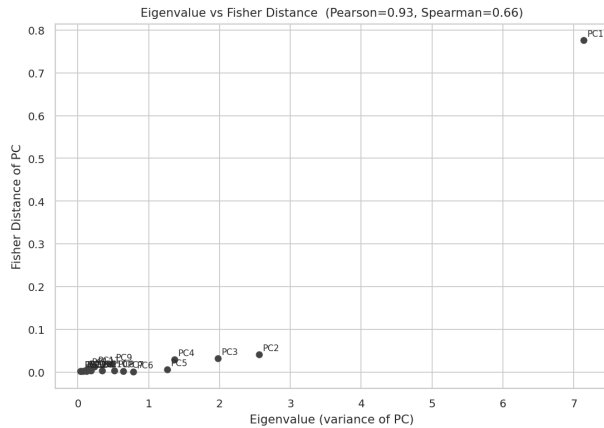


FIGURE 8. Eigenvalue versus Fisher Distance for the 18 principal components. Pearson correlation $r = 0.93$, Spearman $\rho = 0.66$.

Pearson coefficient is driven almost entirely by PC1, which is a strong outlier with both the largest eigenvalue and by far the largest Fisher Distance. The rank correlation $\rho = 0.66$ is more informative: among the remaining PCs, higher-variance components tend to be more class-discriminative, but the ordering is not strict. This is consistent with the theoretical observation that PCA maximizes variance, not class separability; the alignment between the leading PCA direction and the discriminative direction here is a feature of this particular dataset, where the dominant axis of variation is the socioeconomic gradient that drives the crime class.

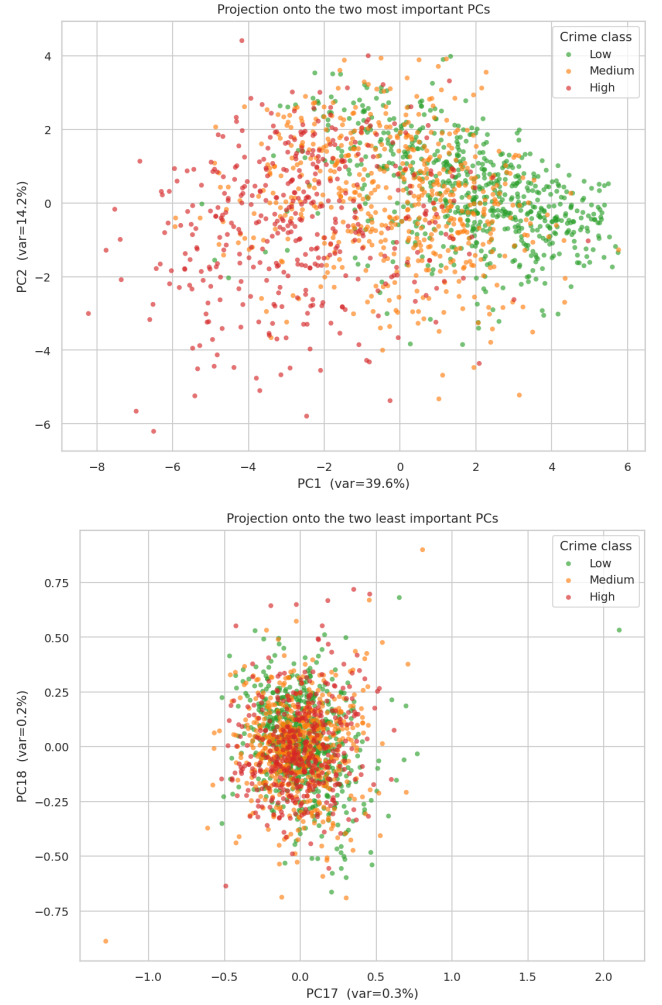


FIGURE 9. Top: projection onto (PC1, PC2). A clear monotone ordering Low $\rightarrow$ Medium $\rightarrow$ High emerges along PC1. Bottom: projection onto the two least important PCs, which show no class structure.

VII. SCATTER PLOTS ON THE FIRST AND LAST TWO PCs (TASK 5)

Figure 9 shows the projection of the cleaned data onto the two most important and the two least important principal components, with samples coloured by their true class label.

The (PC1, PC2) plot shows a clean, almost monotone gradient: *Low*-crime communities concentrate on the negative side of PC1, *High*-crime communities on the positive side, and *Medium* communities form an overlapping band in between. PC2 contributes little additional separation. In contrast, the projection onto the two least important PCs shows essentially no class structure: the three classes are interleaved in a small isotropic cloud, which is expected because these components capture noise-like residual variance.

VIII. LDA PROJECTION AND CLASS-SEPARABILITY COMPARISON (TASK 6)

A Linear Discriminant Analysis was fit with $n_{\text{components}} = 2$. By construction, LDA maximizes between-class scatter rel-

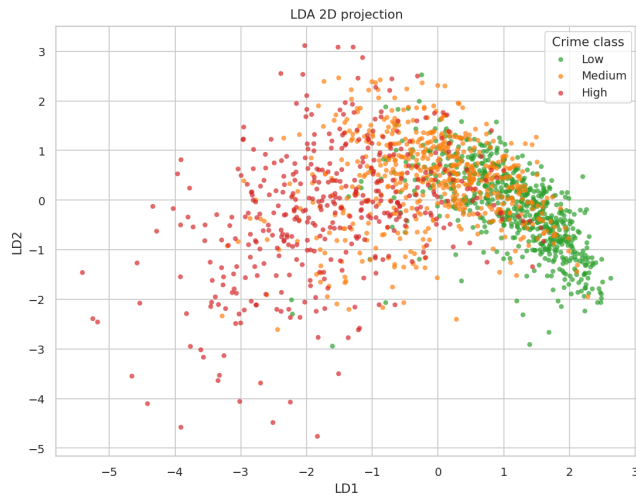


FIGURE 10. LDA 2-D projection. The three classes form a clearer left-right ordering than in the PCA projection.

TABLE 2. Quantitative separability of the 2-D PCA and LDA embeddings. Higher is better in both cases.

Metric	PCA (PC1,PC2)	LDA (LD1,LD2)
Silhouette (true labels)	0.067	0.115
5-NN accuracy (5-fold CV)	0.589	0.611

ative to within-class scatter and therefore tends to produce a tighter class-separable 2-D embedding than PCA when class information is available.

To make the comparison quantitative, two metrics were computed on the 2-D embeddings (Table 2). The silhouette coefficient (with the true class labels as the partition) almost doubles from PCA (0.067) to LDA (0.115). The mean 5-fold cross-validated accuracy of a 5-NN classifier trained on the 2-D embedding also improves from 58.9% on PCA to 61.1% on LDA. Both metrics confirm visually what Figures 9 and 10 suggest: LDA reveals slightly tighter class separability than PCA. The improvement is modest because, as Section VI showed, the leading PCA direction is already strongly aligned with the discriminative direction.

IX. CLUSTERING COMPARISON: K-MEANS, DBSCAN, T-SNE, SOM (TASK 7)

The four algorithms were applied with two different inputs as required by the assignment. K-Means ($k = 3$) and DBSCAN ($\varepsilon = 0.5$, minPts = 10) operate on the 2-D PCA projection (PC1, PC2). t-SNE (perplexity 30, PCA initialization) and a 12×12 MiniSom [2] are fit on the full 18-dimensional z-scored data.

a: K-Means

on the (PC1, PC2) plane recovers three clusters whose boundaries are essentially vertical cuts along PC1. This is consistent with the structure seen in Section VII: PC1 carries almost all of the class signal, so unsupervised partitioning along PC1

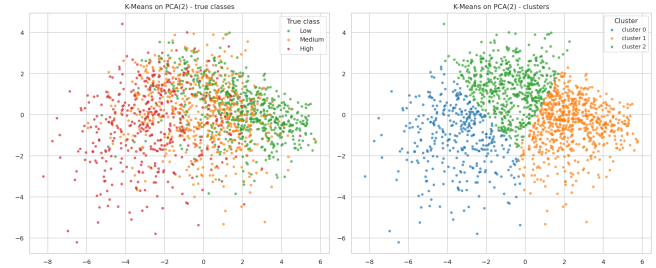


FIGURE 11. K-Means ($k = 3$) on the (PC1, PC2) projection. Left: true classes; right: assigned clusters.



FIGURE 12. DBSCAN ($\varepsilon = 0.5$, minPts = 10) on (PC1, PC2). Most samples form a single dense cluster, with a few small satellites and 149 noise points.



FIGURE 13. t-SNE embedding of the full 18-dimensional z-scored data, coloured by true class. A continuous gradient is visible across the embedding.

TABLE 3. Adjusted Rand Index (ARI) and Normalised Mutual Information (NMI) between cluster assignments and the true class labels.

Method (input)	ARI	NMI
K-Means, $k = 3$ (PCA-2D)	0.191	0.197
DBSCAN ($\varepsilon = 0.5$) (PCA-2D)	0.039	0.061

already gives non-trivial agreement with the true labels (ARI = 0.19, NMI = 0.20). However, the agreement is far from perfect because the three classes overlap heavily along PC2.

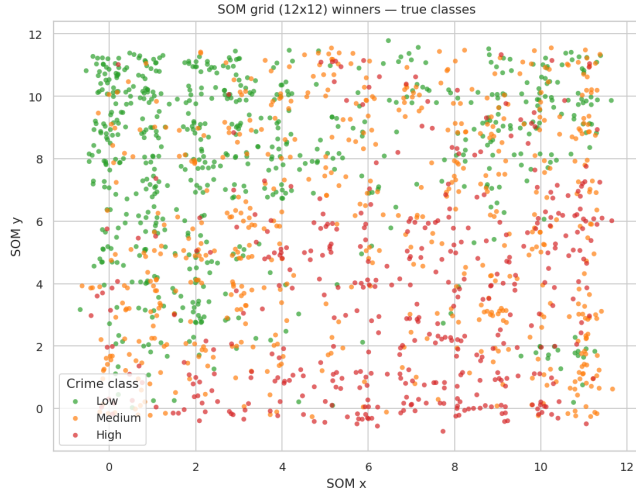


FIGURE 14. Winning units on a 12×12 MiniSom grid trained on the original z-scored data. Small jitter is added so overlapping winners remain visible.

b: DBSCAN

groups the bulk of the PCA cloud into a single dense cluster, isolates a few small satellites, and flags 149 points as noise. Because the data form a single continuous “socioeconomic gradient” rather than well-separated density modes, density-based clustering fails to recover the three classes (ARI = 0.04); DBSCAN exposes that the class structure of this dataset is gradient-like, not modal.

c: t-SNE

on the full 18-dimensional data produces a single, smoothly curved manifold along which the three classes form a continuous gradient (*Low* on one side, *High* on the other). This visualisation reinforces the picture that the classes are largely defined by position along a one-dimensional socioeconomic continuum, not by separate compact clusters.

d: SOM

The 12×12 MiniSom grid spreads samples across the lattice; the colour pattern shows visible class regions, but with substantial mixing. The SOM, being a topology-preserving non-linear projection of the full feature space, confirms the same gradient-like organisation observed by t-SNE.

X. HYPOTHESIS TEST ON PCTKIDS2PAR (TASK 8)

The feature `PctKids2Par` (percentage of kids living in two-parent households) was selected for the hypothesis test because it has the largest Fisher Distance and the largest absolute correlation with the class label (Sections IV–V). The hypothesis to be tested is the equality of its mean between the *Low* and *High* crime classes:

$$H_0 : \mu_{\text{Low}} = \mu_{\text{High}}, \quad H_1 : \mu_{\text{Low}} \neq \mu_{\text{High}}.$$

Two independent random samples of sizes $n = 36$ and $n = 64$ were drawn (without replacement) from each class, and a two-

TABLE 4. Welch’s t-test results on `PctKids2Par` between the Low and High crime classes.

n per class	$\bar{x}_{\text{Low}} (\pm \text{s.d.})$	$\bar{x}_{\text{High}} (\pm \text{s.d.})$	t (p -value)
36	0.795 ± 0.114	0.516 ± 0.143	9.17 (2.0×10^{-13})
64	0.780 ± 0.118	0.512 ± 0.142	11.63 (1.7×10^{-21})
all	—	—	35.27 ($\approx 10^{-162}$)

sided Welch’s t-test (which does not assume equal variances) was applied. The results are summarized in Table 4.

In all three settings (samples of size 36, 64, and the full classes) the p -value is many orders of magnitude smaller than any conventional significance level $\alpha \in \{0.05, 0.01, 0.001\}$. The null hypothesis of equal means is therefore rejected with overwhelming evidence in favour of H_1 : `PctKids2Par` is, on average, substantially higher in Low-crime communities than in High-crime communities (approximately 0.78 vs. 0.51 on the normalized scale). As expected, increasing the sample size from 36 to 64 leaves the means almost unchanged but increases $|t|$, illustrating the higher statistical power of a larger sample for the same effect size.

XI. CONCLUSION

This project applied a complete exploratory and statistical analysis pipeline to the UCI Communities and Crime dataset, with the continuous violent-crime rate discretized into three classes. The following coherent picture emerges. First, family-structure and socioeconomic variables (`PctKids2Par`, `PctIlleg`, `PctPopUnderPov`, `MalePctDivorce`, `racePctWhite`, `medIncome`) carry the strongest individual class signal, as confirmed independently by Pearson correlation and per-feature Fisher Distance, while purely demographic variables such as `pctUrban`, `householdsize`, `agePct12t29`, and `agePct65up` are essentially class-uninformative. Second, the discriminative content is concentrated along a single principal direction: PC1 alone explains 39.6% of the variance and obtains a Fisher Distance (0.78) higher than every original feature except `PctKids2Par`. Eigenvalues and Fisher Distances of principal components are strongly positively correlated ($r = 0.93$, $\rho = 0.66$), but this alignment is a property of this dataset rather than a general rule for PCA. Third, the LDA 2-D embedding marginally outperforms the PCA 2-D embedding in revealing class separability, both visually and quantitatively (silhouette 0.115 vs. 0.067; 5-NN CV accuracy 61.1% vs. 58.9%). Fourth, unsupervised clustering reveals that the class structure of this dataset is gradient-like rather than modal: K-Means partitions PC1 into three slabs and recovers moderate agreement with the labels (ARI 0.19), DBSCAN essentially collapses to a single cluster, and both t-SNE and SOM produce a continuous manifold along which the three classes shade smoothly into each other. Finally, a Welch’s t-test on `PctKids2Par` rejects the null hypothesis of equal class means with extreme significance at both $n = 36$ and $n = 64$, formally confirming the descriptive

evidence of the earlier sections.

Overall, the dataset is best understood not as three well-separated clusters but as a one-dimensional socioeconomic continuum along which violent crime rises monotonically. This explains why simple linear methods (PCA, LDA) capture most of the available structure, why density-based clustering fails, and why the strongest individual predictors are family-structure and deprivation indicators rather than purely demographic ones.

APPENDIX. A

The Jupyter notebook implementing the full analysis pipeline (`Crime_Analysis.ipynb`) and all generated figures accompany this report. The original dataset is publicly available from the UCI Machine Learning Repository [1].

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SERKAN EREN received his B.Sc. degree in Electronics Engineering from Gebze Technical University, Türkiye, in 2023, and is currently pursuing an M.Sc. degree in Artificial Intelligence at the same institution. He works as an Automotive Software Engineer at FEV Türkiye, contributing to Python-based automation, DevOps pipelines, and embedded software integration. His professional interests include deep learning, software engineering for intelligent systems, and the development of data-driven applications for automation and natural language processing.

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